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Computer vision

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1) How does computer vision enables computers :-

It enables computers to capture, analysis, and gives the outcome of the images.

a) Objectives of computer vision :-

→ Object recognition

→ Image processing.

→ analysis and interpret visual scenes.

b) Two real-world applications.

i) health care - x-rays

ii) autonomous vehical - traffic signals

2) Key feature of computer:-

computer vision provides automated images, it is based on AI and machine learning language.

a) Feature extraction is the process of identifying and extracting the characters of the image (i.e. shape, edges, corners etc ...).

b) → Identify and classifies object accurately
→ helps automated decision.
→ Improve image understanding.

3) Different level of image processing in computer vision:-

In computer vision there are three types of level:

- i) low level
- ii) high level
- iii) mid level.

a) low level :-

- > Enhancing the image
- > Noise cancellation.
- > Improving the image.

b) mid - level :-

- > Extracts feature from images
- > Performs segmentation

high - level

- > understanding image
- > Perform object ~~xxx~~ recognition.

Process

Digital image representation :-

Digital image - the computer vision consider the image as digits (numbers).

0's and 1's.

a) Pixel:

One digital image is split into smallest unit, that stores brightness or color information.

b) Higher resolution $\rightarrow$ more pixels $\rightarrow$ better image quality.

lower resolution - Fewer pixels $\rightarrow$ Poor image quality.

5) Image sensing:

Image sensing and acquisition capture light reflected from objects and convert it into digital signals.

a) image sensor:

converts incoming light into electrical signals.

b) Image acquisition device :-

→ Digital camera

→ scanner.

6) Sampling and quantization :-

Both converts the images into digital images.

a) Sampling :-

dividing a image into the pixels.

b) Quantization assigns numeric values to sampled pixels, allow images to be ~~stored~~ in digital.

7) Image formation models :-

It explains how light from a real world objects passes through a camera lens and forms an image.

a) perspective projection:-

Process of projecting a 3D Object onto a 2D image plane.

b) illumination role:-

It provides the light required for capturing images, proper lighting improves.

8) Image processing and computer vision related:-

Image processing improves image quality, while computer vision understands image content and make decisions.

a) Image processing:-

→ Enhancing, restoring or transforming image.

Image processing :-

Enhances
modifies image.

computer vision

understanding
and outcome of
the image.

computer vision is widely used in
health care, transportation, security etc...

a) x-ray, MRI, CT scans to detect
diseases like tumors, fractures.

b) self driving car to detect lanes,
recognize traffic signs.

Digital images fundamental provide
the foundation for storing, processing

a) pixels are the smallest units of
an image, each storing color or
info.

b) Sampling: determines the amount of pixels used to represent the image.

Resolution: determines the amount of details capture.